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Words and deeds – Experimental evidence on leading-by-example

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ABSTRACT

The paper studies how leadership recommendations and leading-by-example shape group behavior. The experimental results from an intergroup contest show that the leaders are the key drivers of contest expenditure. They increase their expenditure substantially once they make recommendations to the followers. The leaders' expenditure recommendations have a highly significant impact on the investments of followers. The additional impact of leading by example confirms the relevance of the alignment of words and deeds. A second experiment replicates the treatments in a non-competitive voluntary contribution game. In this setting, the opportunity for making recommendations has an even stronger impact on the leader's expenditure. For a given level of leader expenditure, the behavior of followers remains remarkably consistent across the two experiments but the additional impact of leading by example is not significant anymore.

Introduction

Leaders encourage followers to contribute towards a common objective. They rally against private opportunism and appeal to common interests, they communicate ambitious targets and give motivational pep talks. However, such statements often fail if leaders do not walk the talk, preaching the proverbial water while drinking wine. Hence, Leading by Example seems to be a particularly powerful method of leadership because actions should enhance the credibility of words (Hermalin, 1998). Battlefields and sport stadiums provide popular case studies for effective exemplary leadership. US-General Patton, for example, is famous for his inspirational address to the Third Army ahead of D-Day in World War II but the speech's impact also relied on Patton's reputation as an audacious soldier (Brighton, 2009). In such situations, it is virtually impossible to disentangle the impact of a leader's words and deeds on the behavior of followers. Hence, I provide clean evidence on the impact of exemplary leadership with an experiment.

The paper studies the impact of Leading-by-Example (LbE) on the voluntary cooperation of followers. It also investigates whether LbE has any additional effect if a 'cheap talk' mechanism (i.e. nonbinding and possibly dishonest communication between players that does not directly affect their monetary benefits) already facilitates intragroup cooperation. The evidence derives from a laboratory experiment in which two groups with three members each compete against each other for a predetermined prize which will be split equally among the winning group's members. The group members can invest resources in order to increase the probability to win a prize. Any increase in investment simultaneously reduces the probability of the opposing group to win the prize in the specific round.

This popular contest model (Abbink, 2010; Dechenaux, Kovenock, & Sheremeta, 2012; Kimbrough, Laughren, & Sheremeta, 2017) provides a stylized version of many real life contests between groups, not just sport events and wars.[1] Patent races illustrate the properties of such a contest. Competing research groups invest time and effort to develop an idea for a new drug or technological improvement. The group that develops it first gets patent coverage and therefore all of the profits from it. The group with the higher investment has better chances to win the race but there is no guarantee for success (Judd, Schmedders, & Yeltekin, 2012). Within each group, a member benefits from the research effort of all other group members. Some input characteristics such as diligence or creativity are difficult to observe, which can give group members an incentive to free ride. They can lazily benefit from the team's performance. It is the role of leadership to ensure that each group member contributes to the common cause. Therefore studying leading-by-example in intergroup contests relates rather closely to leadership problems in commercial enterprises in competitive markets. It also provides an innovative contribution to the experimental literature because economic theory suggests that followers get a monetary benefit from not following their leader (see Predictions section of this paper). Hence the experiment constitutes a particularly critical test for leading by example.

In the baseline treatment of the first experiment on contests, each group member can make a private investment (or contribution) in order to increase her group's chance of winning the prize. In the second treatment, one randomly chosen group member in each group, the

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[1] Note that scholars from outside economics also study closely related contest models. It resembles key features of theoretical conflict models in political sciences (Favretto, 2009; Kydd, 2006; Sambanis & Shayo, 2013).

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'leader', recommends a contribution level before the actual contribution decision. The third treatment allows for Leading by Example. The leader recommends an investment level and decides about her own contribution ahead of the other group members. These 'followers' see both the recommendation and the actual contribution decision of the leader when they decide about their own contribution. In a second experiment, I replicate the three treatments in which group members make a voluntary contribution to a public good. Such a game represents a general cooperation problem in society which manifest itself in tax evasion, abuse of the welfare state or free riding in teams (Gächter & Renner, 2018). This additional experiment allows for a closer comparison of the treatment effects with other experimental studies.

Social scientists, and in particular economists, increasingly use experiments to study leadership in groups (Zehnder, Herz, & Bonardi, 2017). Experiments allow for a random assignment of leadership roles (Arbak & Villeval, 2013; Huck & Rey-Biel, 2006), a controlled variation in the availability of leadership methods and a rather precise measurement of actual behavior and performance. They constitute the methodological "gold standard" in the social sciences (Antonakis, Bendahan, Jacquart, & Lalive, 2010) because they eliminate endogeneity concerns in statistical relationships. Nevertheless experiments are still rather uncommon in the management literature partly because scholars often do not fully appreciate their methodological benefits (Podsakoff & Podsakoff, 2019). Instead most leadership scholars use questionnaires (see Literature section).

Several conceptualizations of leadership consider leaders as providers of role models that foster cooperation in groups. For example, Brown, Treviño, and Harrison (2005) and Walumbwa et al. (2008) claim a predictive validity of integrity in leadership behavior on important work-related attitudes and behaviors. Hence, a leader's behavioral integrity as the perceived alignment of words and deeds (Simons, 2002) plays an important role. However the term *predictive* refers to correlations, which do not establish a causal link. A reverse causality might explain that relationship as well, e.g. if cooperative and dedicated workers induce managers to act in line with their own statements. Moreover, if other factors influence the relationship between motivation and integrity, we have an endogeneity problem and the observed correlation is not informative at all (Antonakis et al., 2010).

Therefore, Walumbwa et al. (2008) call on p 119 for 'experimental designs in which key variables can be manipulated and causality assessed under more tightly controlled circumstances'. This paper responds to this call. The random assignment into leadership roles and the restricted messaging options eliminate the impact of selection processes and leader personality as confounds. The variation of available leadership methods across treatments provides clean evidence on their effect. Here, the separation of statements from observed actions provides a very clean causal identification of the specific role of the element of observable actions in exemplary leadership. Beyond theory testing there is also a policy case for causal identification. An established causal direction establishes a point for policy intervention. We then know that behavioral integrity fosters the commitment of followers. In case of a reversed causality it might be irresponsible to advise leaders on walking the talk if followers remain uncommitted anyway.

The following section explains in greater detail how this experiment contributes to the 'traditional' leadership literature as well as to behavioral economics. Design of the intergroup contest experiment section introduces the experimental design for the contest experiment, Predictions section the hypotheses and Results section the results. In Replication of the treatments with a non-competitive game section I present the design and results of the experiment on leading by example in a noncompetitive environment. The last section summarizes the findings and provides both scientific and normative conclusions.

Literature review

Several leadership concepts claim a positive impact of exemplary leadership on follower behavior. Podsakoff, MacKenzie, Moorman, and Fetter (1990) argue that '*transformational leader behaviors influence extra-role or organizational citizenship behaviors*' (p. 109). Avolio (1999) claims on p 43 that transformational leaders provide '*role models for followers to emulate (…) and display high standards of ethical and moral conduct*'. This claim suggests a mechanism like how leading by example influences follower behavior. If leaders act as they preach it increases their perceived integrity and trustworthiness (Bass & Steidlmeier, 1999). In consequence, Simons (2002) conceptualize the antecedents and consequences of behavioral integrity of managers. Behavioral antecedents establish a pattern of word-deed alignment. The employees' perception of this pattern forms the manager's integrity. The integrity shapes the employees' trust and subsequent behavior. Simons (2002) emphasizes the roles of perceptual processes and biases that can establish a mismatch between a leader's intentions and the followers' perceptions of leader integrity. Note that Palanski and Yammarino (2007) provide an overview of different, and overlapping, concepts that include integrity. Furthermore, economic theory also suggests positive benefits of a word-deeds alignment because it should provide more credible signals about a leader's intentions and private information (Hermalin, 1998).

All these conceptual contributions fostered a wave of empirical studies. Simons, Leroy, Collewaert, and Masschelein (2015) summarize the relevant results in their meta-analysis on the behavioral integrity of leaders. They find that behavioral integrity has rather high correlations with trust, in-role task performance and organizational citizenship behavior. The analysis relied on 35 samples from published and unpublished studies that report the correlation between leader behavioral integrity and follower consequences. Similarly Bedi, Alpaslan, and Green (2016) as well as Zhang et al. (2019) use meta-analyses to identify the impact of ethical leadership. As far as I could identify, all samples used questionnaires (see also Piccolo, Buengeler, and Judge (2018)). Only one, subsequent, study employed a setup with some experimental variation in this context. Friedman et al. (2018) studied three scenario-based experiments that look at response to word–deed inconsistency in different cultures. The studies in that paper consisted of a set of experimentally manipulated business scenarios in which promises were (assumed to be) either kept or broken. However, the subsequent questionnaires did not elicit an actual or hypothetical behavioral response to those scenarios (i.e. an immediate consequence of behavioral integrity) but rather an assessment of the scenarios regarding the perceived integrity. Hence, even in this setting, we do not get a causal identification regarding the impact of behavioral integrity. This paper addresses the deficit. It provides, in a consequential and incentivized setting, a causal identification regarding the impact of a leader's word-deeds alignment on the subsequent behavior of the followers.

A well designed experimental study eliminates endogeneity concerns that restrict the causal interpretation of correlational effects in most questionnaire studies (Antonakis et al., 2010). An experiment also circumvents related problems of selectivity. Few organizations establish random team formation and leadership appointment. Furthermore, leaders do not deploy leadership methods arbitrarily. Hence, a questionnaire study cannot isolate the impact of a leadership method from associated team and leader characteristics. Some clever studies use random variation in field data to analyze natural experiments. For example, Besley, Montalvo, and Reynal-Querol (2011) study random leadership transitions because of natural death or terminal illness to show that political leaders matter for growth. Questionnaires are also a poor substitute for direct independent observations of a behavior. Questionnaire responses about leader and follower behavior are clearly judgmental and require additional information about whether these judgements are correct.

Furthermore, there is a specific case for experiments in the context of a leader's behavioral integrity. The focus of behavioral integrity is on the word-deeds alignment (Simons, 2002). Real-life observations and questionnaires cannot disentangle words from deeds. Actions and statements interact and often occur simultaneously. Hence it is impossible to assess the additional impact of the (non-)aligned behavior on top of the effect of verbal statements. The emphasis on perceptional processes in Simons

(2002) indicates that purely observational data are confounded. These problems call for a targeted experimental study. Podsakoff and Podsakoff (2019) provide a very helpful discussion of the strengths and limitations of experimental designs in leadership research in this context.

Over the last two decades behavioral economics has provided several experimental studies to address this deficit in leadership research. This branch of economics with multiple interdisciplinary connections investigates systematic deviations from the homo oeconomicus model of selfish, rational agents. Several theoretical contributions provide testable hypotheses about specific social motives, for example the perceived altruism of others (Levine, 1998), inequity aversion (Fehr and Schmidt, 1999) or inferred intentions (Falk & Fischbacher, 2006). Some models are close to contributions from psychology and management studies. As in Simons' (2002) concept of behavioral integrity, reciprocity and trust are important factors in behavioral economics (Berg, Dickhaut, & McCabe, 1995; Fischbacher & Gächter, 2010; Fischbacher, Gächter, & Fehr, 2001). People contribute to a common good if they expect other people to do so as well. Hence, a leader can actually foster contributions of followers because some people will respond in kind to a contributing leader.

Several experimental papers study how leaders promote such voluntary cooperation in noncompetitive environments (e.g. Gächter, Nosenzo, Renner, & Sefton, 2013; Güth, Levati, Sutter, & Van Der Heijden, 2007; Levati, Sutter, & Van der Heijden, 2007). They reveal that groups perform best when led by those who are cooperatively inclined. Drouvelis and Nosenzo (2013) find that a shared group identity enhances this effect. Cartwright, Gillet, and Van Vugt (2013) and Gächter and Renner (2018) identify how leaders shape the beliefs of followers in this context. Note also that the literature does not provide many experimental comparison between verbal statements and actual leadership behavior. Pogrebna, Krantz, Schade, and Keser (2011) observe that non-binding verbal statements of a leader have a similar effect like Leading-by-Example whereas Dannenberg (2015) reports a relatively low effect of such chat messages. Contrarily, Koukoumelis, Levati, and Weisser (2012) show that one-way communication has a beneficial impact on cooperation. Dal Bó and Dal Bó (2014) observe that a non-binding message with a moral standard leads to a significant but transitory increase in cooperation.

Rather little experimental research focuses on how a group's exposure to competition shapes the leader-follower relationship, or, more precisely, how it influences the impact of leadership behavior on the decisions of followers. In his doctoral thesis, Heine (2017) reports results from an intergroup contest with leadership. He finds that leadership prompts an escalation of the contest. The results in a working paper of Weisser (2011) indicate that both leading by example and intergroup competition significantly increase contributions in a public good game, whereas the interaction between leading by example and intergroup competition remains insignificant. Unlike other social dilemma situations, intergroup contests with predetermined prizes generate an incentive to invest some resources for the benefit of the own group (Abbink, 2010; Abbink, Brandts, Herrmann, & Orzen, 2010). It is well known from social psychology and behavioral economics that a group's exposure to competition has a significant impact on its strategy and its internal culture (Bornstein, Gneezy, & Nagel, 2002; Goette, Huffman, Meier, & Sutter, 2012; Kosfeld & von Siemens, 2011). Group members generate two types of external effects on the monetary payoffs of the other participants if they invest in intergroup contests. They generate a benefit to their fellow group members and simultaneously harm the members of the opposing group because the other group has a reduced chance to win the prize. This negative external effect can foster a preference for winning (Mago, Samak, & Sheremeta, 2016; Sheremeta, 2013) or parochial altruism, when prosocial behavior towards one's own group goes along with hostility towards the opposing group (Abbink, Brandts, Herrmann, & Orzen, 2012). In consequence, contestants make rather large contributions to improve the odds of their own side (Sheremeta, 2015). Because of such seemingly excessive investments, individuals often choose to employ competition between groups in order to promote within-group cooperation (Markussen, Reuben, & Tyran, 2014; Puurtinen, Heap, & Mappes, 2015).

Design of the intergroup contest experiment

In the experiment two groups competed against each other for a predetermined prize. The experiment lasted for 20 rounds, each round the group faced a randomly chosen opponent. A group consisted of three players, one leader and two followers, who kept their roles throughout the entire experiment.[2] Note that the opponents changed across rounds. To gain more independent observations, each session was divided into two different matching groups of four groups each. A matching group is a group of four groups that interact with each other over time but not with the other groups in that particular session.

Within a matching group the individuals groups are randomly matched as contestants in each period. Because of anonymous interaction the groups never know the specific opponent in a given round.[3] Imprecise coding prevented an exact pair-wise matching of the groups. More specifically, in each round three groups in a matching group were compared to the fourth group, while that specific group was compared to one of the other three groups. This benchmark group changed randomly across rounds. Because of the anonymous matching the participants did not realize this unintended outcome in any of the treatments.[4] Table 1 summarizes the sequence of the stage game and the treatment differences. The game did not vary across the rounds in a specific treatment.

I now explain the game in a round in the Baseline Treatment in greater detail. Afterwards, I describe the differences between the treatments.

Baseline treatment

Each player received an endowment of 100 points in a round, with 120 points exchanging into 1 €. The player could keep the points in a private account or invest them in order to enhance the chances of the own group in the intergroup contest. The groups compete for a prize of 300 points. Irrespective of the individual investments, each group member receives 100 points if her group wins. As in benchmark contest design of Tullock (1980) the winning probability of a group is determined by its members' investments relative to total investment in both groups. At the end of a round, the followers learn about their own payoff including whether the own group has won the prize. The leaders also receive information about the individual investments of group members and the total investment of the opposing group. Such a Tullock contest is arguably the most widely used model for the experimental study of conflicts (Abbink, 2010; Dechenaux et al., 2012; Kimbrough et al., 2017).

Recommendation treatment

In the Recommendation Treatment the leader makes a non-binding investment recommendation to her followers. All group members see this recommendation when they make their subsequent investment decision.

Leading-by-example treatment

In the Leading-by-Example Treatment the leader also makes a non-binding investment recommendation to her followers. Afterwards the leader makes her own investment decisions. The followers see the

[2] Appendix A provides an English translation of the experimental instructions (original in German). The instructions did not mention terms like leadership or followers and used neutral labels, referring to players A, B and C (the leader). This procedure should minimize demand effects that beset many applied experimental studies (Zizzo, 2010). I used a partner-matching protocol which implies that the group composition does not change across rounds.

[3] Baik, Chowdhury, and Ramalingam (2015) explore the relevance of such matching rules in intergroup contests. There results suggest that contests without such stranger matching facilitate tacit collusion between the opponents.

[4] Neither the post-experimental questionnaire nor verbal reports from the research assistants indicate that any of the participants has observed the problem. The author of this paper is solely responsible for the error.

Table 1
The design of a round in the experiment.

Step	Experimental treatments		
	Baseline	Recommendation	Leading by example
0	Random matching of opposing groups		
1	Each player gets an endowment of 100 points		
1a		Leader makes a non-binding investment recommendation	
1b			Leader decides about investment
2		Followers learn about the recommendation	Followers learn about the recommendation and the leader's investment.
	Leader and followers decide about investment		Followers decide about investment
3	All players learn whether the group has won the prize. Leaders learn about the individual investments of the followers and the aggregate investments of the other group.		

recommendation and the actual contest expenditure of the leader when they make their subsequent investment decision.

Predictions

In this Introduction section present two sets of competing hypotheses. The first set derives from standard game theory[5] assuming common knowledge that all actors are rational and want to maximize their own payoff. The stage game is a contest between two groups, 1 and 2. Each group includes a leader (L_i, with $i \in \{1;2\}$ denoting the respective group) and two followers (F_{ia} and F_{ib}). The groups compete for a prize $P > 0$, the winning groups divides the prize in equal shares among its three members. Each group member receives an endowment $X > 0$ and can spend $0 \leq x \leq X$ to enhance the winning probability of her group. The winning probability of a group in Tullock contest is determined by the members' investments relative to total investment in both groups (i.e. the sum of all individual investments).

Now I study the payoff function of a risk-neutral follower in group 1. The payoff increases in the endowment X as well as the product of the winning probability and the share of the prize P. It decreases in the contest expenditure. The follower chooses the expenditure to maximize the expected payoff

$$\max_{x_{F1a}} \frac{X_1}{X_1 + X_2}\left(\frac{P}{3}\right) - x_{F1a},$$

with $X_1 = x_{L1} + x_{F1a} + x_{F1b}$ denoting the investments of leaders and followers in the first group and $X_2 = x_{L2} + x_{F2a} + x_{F2b}$ for the second group. The maximization of the aforementioned function with respect to x_{F1a} yields the following optimality condition:

$$\frac{X_2}{(X_1 + X_2)^2}\left(\frac{P}{3}\right) - 1 = 0$$

A transformation of that condition implies $\frac{X_2}{(X_1+X_2)^2}\left(\frac{P}{3}\right) = 1$, and accordingly for the other group, $\frac{X_1}{(X_1+X_2)^2}\left(\frac{P}{3}\right) = 1$. Taken together, the two conditions imply that both groups make the same total expenditure ($X_1 = X_2$). The resulting simplification of the optimality condition gives us

$$\frac{P}{12} = X_1$$

Now, because of $X_1 = x_{L1} + x_{F1a} + x_{F1b}$, we obtain the investment of the follower

$$\frac{P}{12} - x_{L1} - x_{F1b} = x_{F1a}$$

This result shows that the investments of the group members are strategic substitutes. If one group member decreases her investment she increases the investment incentives of the other group members, and vice versa, accordingly $\left(\frac{\partial x_{F1a}}{\partial x_{L1}} = -1\right)$. This analysis leads to an incentive against leading by example. At the group level, there is an optimal best response to the behavior of the opposing group. Whatever the leader contributes to this best response does not have to be provided by the followers. If the leader does not provide any investments, the followers make all the investments themselves. Followers should reduce their investments if they see a high investment of the leader. Note also, that the leader's recommendations are actually irrelevant in this context. As a result, we get multiple equilibria within groups. Each group has an optimal overall strategy but there are multiple combinations of individual decisions that could implement this strategy. More specifically, all these potential combinations share one crucial characteristic, namely the total investment of all group members is $x_{L1} + x_{F1a} + x_{F1b} = \frac{P}{12}$. With $P = 300$ points, the predicted *total* (or aggregate) group expenditure is 25 points.

Hypothesis 1. (common knowledge about payoff maximizing preferences):

a) Total group expenditure does not differ across the treatments.
b) In the Baseline and the Recommendations Treatments, the investments of leaders and followers do not correlate with each other.
c) In the Leading by Example Treatment the investments of the followers ***decrease*** with increasing investments of the leader. Leaders will invest less than followers in this treatment.

It is noteworthy that total expenditure of 25 points is suboptimal from a social welfare perspective. The contest expenditure does not increase the prize. Hence, it generates no value. From an efficiency point of view, both groups should coordinate on a zero expenditure strategy. The total expenditure is also suboptimal if one focuses the expected total payoffs of the individual groups. In such a context, the contest expenditure is too low. The term $\left(\frac{P}{3}\right)$ in the payoff function implies that the group members only take their own private payments into account but not the benefit of the entire group (P). Hence, with common knowledge, that all participants want to maximize the payoffs of their respective groups, the total expenditure should accumulate to 75 points rather than 25.

Hypothesis 1 is not in line with well-observed behavioral patterns of leadership and competitive behavior in groups. First, people often make seemingly excessive investments in competitive environments (Abbink et al., 2010; Abbink et al., 2012; Mago et al., 2016; Sheremeta, 2015). This observation suggests that the total expenditure in a group exceeds the aforementioned 25 points. Second, abundant evidence revealed that most people are conditionally cooperative. They contribute to a common cause if others do so as well (Fischbacher et al., 2001; Fischbacher & Gächter, 2010). Hence, a leader can actually foster contributions of followers because some people will respond in kind to a contributing leader. As Kosfeld (in press) points out, leaders signal their belief about how they expect the followers to behave. In the context of this experiment they have two methods to do so, the nonbinding recommendation and their actual contribution (in the LbE Treatment). LbE provides a stronger signal because the followers know whether the leader actually cooperates. They can assess the leader's behavioral integrity and the leaders should be aware of this (Simons, 2002). In consequence, groups with investing leaders in the LbE Treatment should see the highest contributions.

Hypothesis 2.

a) Total contest expenditure per group i is ordered as follows across the treatments:

$$25 < X_{i,Base} < X_{i,Rec} < X_{i,LbE}$$

b) The correlations between the investments of leaders and followers increase across the treatments in the following order

$$0 = corr(x_{L1}, x_{F1})_{Base} < corr(x_{L1}, x_{F1})_{Rec} < corr(x_{L1}, x_{F1})_{LbE}$$

[5] Game theory provides models of strategic interaction in between rational and selfish decision-makers.

Results

Procedure

The data were gathered in nine sessions, three for each treatment, with altogether 216 subjects. All sessions took place in autumn 2016 at the Lakelab at the University of Konstanz. Subjects were students from the University of Konstanz who were recruited with the software "ORSEE" (Greiner, 2015). The experiments were computerized with the software "z-Tree" (Fischbacher, 2007). Each subject participated in one of the treatments only. They received written instructions and comprehension questions that had to be answered correctly before the experiment could start. An English translation of the instructions is included in Appendix A of this paper. The sessions lasted approximately 50 min and subjects received on average about 19 Euro (about 21 US Dollars in 2016). All subjects received their payment privately at the end of their session.

Descriptive statistics

Table 2 shows the leaders' average investment recommendations and the actual contest expenditure of both leaders and followers in the three treatments across the 20 rounds. The table also shows group expenditure. The following comparisons rely on group averages across rounds, using Wilcoxon rank-sum and signed-rank tests.

The main impression is that leaders react more strongly to the treatment differences than followers do. The leaders' investments are significantly higher in the Recommendations and Leading-by-Examples treatments than in Baseline ($p = .0304$ and .0126). Leaders do not invest significantly less than followers in the Baseline Treatment ($p = .1615$, according to the Wilcoxon signed-rank test) but significantly more if they lead by example ($p = .0199$, according to the Wilcoxon signed-rank test). Meanwhile, the follower's investments are somewhat higher in the Recommendations Treatment (the impact is marginally significant at $p = .0526$, according to the Wilcoxon rank-sum test) but they are not higher in the LbE Treatment when they also observe the leader's actual investment decision.

Furthermore leaders inflate recommendations if they do not have to reveal their own decisions. The leaders' recommendations are somewhat higher in the Recommendations Treatment than in then Leading-by-Example Treatment (the impact is marginally significant at $p = .0833$, according to the rank-sum test). They are also significantly higher than leader efforts in that treatment (Wilcoxon signed-rank test, $p < .0001$) whereas they are not in the LbE Treatment.

The normative implications are quite interesting. The total investments (i.e. the sum of individual investments of all group members) do not just exceed the game theoretic predictions (25 points) but also the optimal investment level of 75 if all members focused exclusively on the benefit of their own group. ($p = .0223$ in the Baseline Treatment, $p = .0012$ in the Recommendations Treatment and $p = .0012$ in the LbE Treatment, according to the Wilcoxon signed-rank test). Hence the participants appear as if they want to win almost irrespectively of the associated cost, and leadership enforces this preference.

Table 3 shows the average payoffs of leaders and followers across treatments, measured in points (120 points buy 1 €). The payoffs are the highest in the Baseline treatment with the lowest expenditure. Since contest expenditure does not add value, a higher expenditure leads to lower average payoffs. For benchmarking, fully cooperative behavior among all contestants would yield 150 points per player per period whereas complete conflict expenditure would leave only the half the prize as expected payoff (i.e. 50 points).

Leaders

Now I study the treatment differences in greater detail. The first focus is on the investment of the leaders. The estimations in the first two models in Table 4 confirm the significantly lower investments in the Baseline Treatment. The Recommendations Treatment serves as a benchmark in these models. Model 2 includes additional control variables which increase the explained variance but have no qualitative impact on the treatment comparisons. They reveal a significant decline in investment over time (variable Round *t*, which ranges from 1 to 20, see also Fig. 1) and a strong persistence of behavior across rounds (Investment t_{-1}). Model 3 estimates the statistical relationship between the leaders' recommendations and their own investments in LbE and Recommendation Treatments. This relationship is highly significant. Interestingly, it does not differ between the two treatments even though the followers cannot identify in the Recommendations Treatment whether the Leader sticks to her own advice.

Followers

In Table 2, the seemingly small differences of the followers' investments across the treatments suggested that leadership has a rather limited impact in the experiment. However, an analysis of the behavioral relationship within groups reveals a different insight. Table 5 shows an increase in the correlation between leader and follower investments if leaders can make a recommendation. This correlation increases again if the leaders lead by example. Unsurprisingly, the followers focus their decision more strongly on the leader's investment than her recommendation.

Table 6 explores these correlations in greater detail. Its models estimate the investments of the followers and their statistical relationship with the investments of the leader across the treatments. Again, the Recommendations Treatment serves as reference. Model 1 shows that all treatment differences are insignificant if we control for the investment of the leader. Model 2 compares the statistical relationship between the investments of the leader and the followers across the treatments. This relationship is at its lowest in the Baseline Treatment (in which it is insignificant), increases in the Recommendation Treatments (in which the leaders recommendations provide a causal link between the two investments) and has the highest coefficient in the Leading by Example Treatment where one can claim a clear causal impact of leadership investments on the contributions of the followers. The negative coefficient for the main effect (*variable Leading by Example Treatment*) explains why the enhanced coordination within groups with leading by example does not translate into higher aggregate cooperation. Model 3 includes additional control variables. The coefficients regarding the statistical relationship of the investments in the different treatments become smaller but remain significant.

The aforementioned results contradict some key predictions derived from economic theory. In particular, they show that the investments of

Table 2
Average investment recommendations and actual contest expenditure of leaders and followers across rounds (standard deviations in parenthesis).

	Baseline	Recommendations	Leading by example (LbE)
Leader recommendations	–	54.28 (14.82)	46.06 (16.44)
Leader investments	27.29 (19.61)	41.17 (17.33)	39.43 (16.13)
Follower investments	33.63 (13.50)	42.04 (16.32)	36.31 (14.88)
Total group investments[a]	94.56 (49.11)	125.25 (65.04)	112.06 (71.65)

N per cell: 24 Leaders/48 Followers in 24 Groups.

[a] Total group investment is the sum of the investments of the three group members.

Table 3
Payoffs of leaders and followers per period (in points, with 120 points exchanging into 1€).

	Baseline	Recommendations	Leading by example (LbE)
Payoff leader	126.04 (55.15)	110.70 (48.83)	110.98 (47.96)
Payoff follower	119.70 (52.82)	109.84 (50.30)	114.10 (49.82)

N per cell: 24 Leaders/48 Followers in 24 Groups.

leaders and followers are strategic complements, not substitutes. If the leader raises her contributions, the followers do so as well. As Hypothesis 2 predicts, most followers are conditionally cooperative. They raise their investments when the leaders do so as well. We also observe some differences in investment levels across the treatments. However, the leaders are the main drivers of this effect. They tend to underinvest in the Baseline Treatment but increase their expenditure substantially once they make recommendations to the followers. Leading by example, the visible ex-ante investment commitment of the leader, has no discernible impact on the investment level and a small positive impact on followership. This result provides support for the relevance of behavioral integrity.

Replication of the treatments with a non-competitive game

The previous design relates rather closely to leadership problems in commercial enterprises in competitive markets. I replicate the three treatments in a second experiment in which groups do not compete with each other. This additional experiment allows for a closer comparison with other experimental studies. It incorporates a non-competitive voluntary contribution mechanism (VCM) in groups as it has been used in the experimental literature on leadership so far (Gächter et al., 2013; Güth et al., 2007; Levati et al., 2007). The game captures a basic social dilemma of many groups in which individual group members do not have an incentive to contribute (efficiently) to the common good. Corporate governance in a firm with dispersed ownership provides an example for such a dilemma (Shleifer & Vishny, 1997). All shareholders benefit from improved managerial oversight. However, the benefit for an individual small shareholder is rather small. Hence there is little incentive to provide the time and effort required to check the decisions of the firm's management. Each shareholder hopes that others will put in the required resources. The resulting underinvestment in corporate governance does not take the benefit of individual effort on the fellow shareholders into account.

Design and procedure

This second experiment also lasted for 20 rounds. Again, each group consists of three randomly chosen players, one leader and two followers who kept their roles throughout the entire experiment. Each player received an endowment of 100 points in a round, with 120 points exchanging into 1 €. The player could keep the points in a private account or invest them in order to enhance the welfare of all group members. More specifically, one point invested implied a marginal per-capita return for each group member of 0.5 points and a total return of 1.5 points. Hence, it is efficient to invest all points but an individual who focuses just on her own payoff will not invest any point. The Recommendation and the Leading by Example Treatments proceeded as described above. At the end of a round, the followers learned about their own payoff. The leaders also received information about the individual investments of group members. This second experiment was conducted also at the University of Konstanz in May and June 2016 with 120 students from the same subject pool. The smaller sample size also reflects that each group constitutes an independent observation in this experiment. The procedural aspects were the same as in the first experiments.

Descriptive statistics

Table 7 summarizes the results of leadership in the voluntary contribution mechanism. The Recommendations and the Leading by Example Treatments see a particularly strong increase in investments. The followers' investments are higher in Recommendations and LbE than in baseline ($p = .0048$ and $p = .0465$, respectively; see also Fig. 2), and so are the investments of the leaders ($p = .0008$ and $p = .0027$, respectively). Interestingly, recommendations are significantly higher than leader efforts in both treatments ($p = .0117$ and .0295, respectively). Because the total welfare increases in the investments, the treatment modifications have a benign impact.

Leaders

The estimations documented in Table 8 show that the Leaders' investments increase once they have to make recommendations to the followers (Model 1). The investments also correlate significantly with their own recommendations irrespective of whether the followers are informed about their leader's investments or not. (Model 2) Again, we can observe a significant decline in investment over time (Round *t*) and a strong persistence of behavior across rounds (Investment t_{-1}).

Followers

Table 9 now studies the relationship between the investments of leaders and followers in this second experiment in greater detail. Like

Table 4
OLS estimations of treatment differences in leader investments (with recommendations treatment as reference).

Dep. var: Investment leader	Model 1		Model 2		Model 3 (w/o baseline)	
Baseline treatment	−13.881**	(5.037)	−6.107**	(2.784)		
Leading by example treatment	−1.742	(4.554)	−0.861	(2.460)	2.030	(5.028)
Leader recommendation					0.481***	(0.062)
LbE × recommendations					0.021	(0.128)
Round *t*			−0.532**	(0.100)	−0.359***	(0.089)
Winner t_{-1}			−0.923	(1.211)	−0.268	(1.205)
Investment t_{-1}			0.547***	(0.061)	0.390***	(0.105)
Constant	41.173***	(3.249)	24.407***	(4.164)	2.483	(4.154)
R^2	0.051		0.353		0.525	
N (obs/matching group)	1440/18		1368/18		912/12	

In parentheses: standard errors clustered at the matching group level. The variable *Round t* has a range from 1 to 20, given the actual round in which the investment has been made, Each matching group included four groups that were randomly matched into contests across the rounds. ***: $p < .01$; **: $p < .05$; *: $p < .1$.

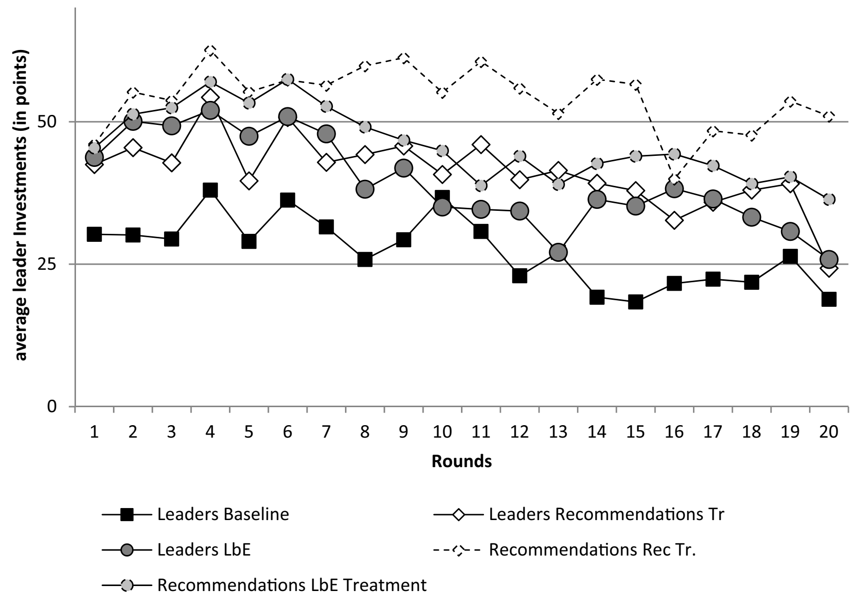

Fig. 1. Investments and recommendations of leaders across treatments and over time.

Table 5
Correlations between recommendations and investments across the treatments.

Baseline treatment	Leader investments	Follower investments
Leader investments	1	0.122
Follower investments		1

Recommendations treatment	Leader recommendations	Leader investments	Follower investments
Leader recommendations	1	0.576	0.548
Leader investments		1	0.437
Follower investments			1

Leading by example treatment	Leader recommendations	Leader investments	Follower investments
Leader recommendations	1	0.652	0.606
Leader investments		1	0.682
Follower investments			1

models 2 and 3 in Table 6, it estimates the investments of the followers and their statistical relationship with the investments of the leaders across the treatments. The Recommendations Treatment serves as a benchmark. Model 1 shows the significant relationship between the investments of the leader and the followers. This relationship is significantly stronger than in the Baseline Treatment but not significantly weaker than in the Leading by Example Treatment. Model 2 includes the aforementioned control variables which increase the explained variance but have little qualitative impact on the results.

Discussions

This paper studied whether Leading by Example improves cooperation and coordination in groups if leaders can already use nonbinding and costless signals (i.e. cheap talk). The experimental setup focused on group behavior in intergroup contests. A second experiment replicated the experimental treatments in a non-competitive environment. In both cases the results do not find a significant *additional* benefit of exemplary leadership in terms of total payoffs. Leading-by-example increases followership significantly in the contests. Leaders and followers make more coherent choices when the latter can see the decisions of the former. In the non-competitive environment the additional effect of leading-by-example points in the same direction but it is insignificant.

The paper shows how experimental research with a background in behavioral economics can enrich leadership conceptually and methodologically. Concepts like behavioral integrity (Simons, 2002) overlap with economic theories of prosocial behavior that deviate from homo economicus assumptions and take reciprocity into account (e.g. Falk & Fischbacher,

Table 6
OLS estimations on the relationship between the investments of leaders and followers (with recommendations treatment as reference treatment).

Dep. var: Investment follower	Model 1		Model 2		Model 3	
Investment leader	0.548***	(0.114)	0.529***	(0.124)	0.310***	(0.080)
Baseline treatment	−0.794	(4.646)	7.112	(8.548)	8.104*	(3.860)
Baseline × investment leader			−0.299*	(0.161)	−0.273***	(0.078)
Leading by example treatm.	−4.770	(4.045)	−18.808**	(7.857)	−7.950**	(3.423)
LbE × investment leader			0.355**	(0.152)	0.140*	(0.073)
Round t (1 − 20)					−0.148*	(0.079)
Winner t_{-1}					−1.122	(1.303)
Investment t_{-1}					0.591***	(0.044)
Constant	19.470***	(5.648)	20.240***	(6.615)	6.610**	(3.411)
R^2	0.097		0.117		0.430	
N (obs/matching group)	2880/18		2880/18		2736/72	

In parentheses: standard errors clustered at the matching group level. Each matching group included four groups that were randomly matched into contests across the rounds. ***: $p < .01$; **: $p < .05$; *: $p < .1$.

Table 7
Investment recommendations and actual expenditure of leaders and followers in the VCM (standard deviations in parenthesis).

	Baseline	Recommendations	Leading by example (LbE)
Leader recommendations	–	81.66 (13.89)	73.34 (21.84)
Leader investments	27.56 (14.94)	65.47 (26.59)	67.33 (25.75)
Follower investments	31.41 (14.00)	65.40 (25.50)	57.35 (30.44)
Total group investments	90.33 (37.68)	196.26 (78.30)	182.04 (88.49)
N (leader & groups/follower)	16/32	8/16	8/16

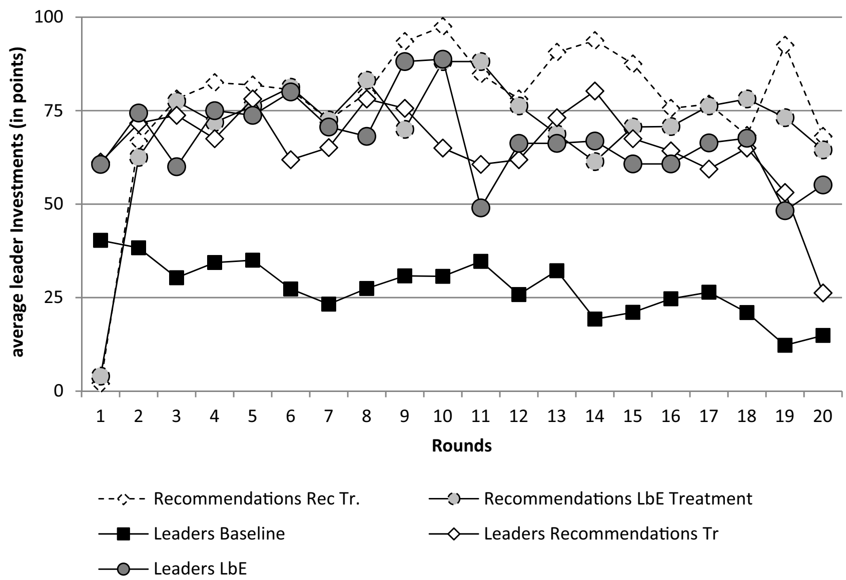

Fig. 2. Investments and recommendations of leaders across treatments and over time in the non-competitive VCM.

Table 8
Leader investments in the VCM (with recommendations treatment as reference point).

Dep. var: Investment leader	Model 1		Model 2	
Baseline treatment	−14.920***	(5.305)		
Leading by example treatment (LbE)	1.725	(5.065)	15.614	(11.777)
Leader recommendation			0.607***	(0.186)
LbE × recommendation			−0.117	(0.189)
Round t (1–20)	−0.569***	(0.134)	−0.643***	(0.203)
Investment t_{-1}	0.608***	(0.073)	0.442***	(0.134)
Constant	30.879***	(7.616)	−7.086	(5.560)
R^2	0.576		0.627	
N (obs/matching group)	608/32		304/16	

In parentheses: standard errors clustered at the matching group level. In the first experiment each matching group included four groups that were randomly matched into contests across the rounds. In the second, each group constituted a matching group because the experiment does not allow for interaction between groups. ***: p < .01; **: p < .05; *: p < .1.

2006). Behavioral integrity focuses on the importance of the word-deeds-alignment in managerial actions because such an alignment fosters trust among followers. The experiments reported in this paper disentangled the additional impact of the deeds on top of the words, which is virtually impossible with real world observations and questionnaire studies.

It is surprising that leading by example does not improve economic outcomes in comparison to simple recommendations. However, this observation does not entirely invalidate the claim for behavioral integrity. The results from the first experiment reveal a more differentiated behavioral pattern. Cheap talk leads to rather inflated recommendations. Both leaders and followers deviate from these recommendations but they coordinate their actions significantly better than in the Baseline Treatment without recommendations. The correlation between the decisions is much stronger. This coordination increases even more if followers see the leader's recommendation and action which is in line with behavioral integrity. Leaders just reduce their recommendations because they do not want to make high contributions themselves. This behavior is actually rather sensible because the groups are often too competitive.

Limitations and suggestions for future research

This paper provides a novel contribution on the relevance of the work-deeds alignment in leadership. Because its empirical analysis relies on results from an abstract lab experiment conducted with university students its external validity is rather limited (Galizzi & Navarro-Martínez, 2018). The internal validity is clearly the strong point of the paper. Hence, it provides a complementary contribution to other empirical studies that could not establish causal relationships between different factors of leadership theory.

Some readers may also question specific features the design. However, because variability and replicability are great benefits of experiments, these concerns should foster additional experimental studies. Most practitioners would probably call for a less parsimonious and more intuitive work environment that takes more factors into account. This concern induced a strong growth of field experiments (Harrison & List, 2004; Levitt & List, 2009; List, 2011).[6] More realistic work environments typically generate problems of measuring behavior and performance and assigning it to individual agents. The literature has provided several standardized procedures for disentangling social and psychological processes in more complex economic interactions

[6] The American Economic Association's registry for randomized controlled trials and field experiments lists all current and past studies at www.socialscienceregistry.org

Table 9
OLS estimations on the relationship between the investments of leaders and followers in the VCM (with recomm. treatment as reference point).

Dep. var: Investment follower	Model 1		Model 2	
Investment leader	0.673***	(0.146)	0.360***	(0.088)
Baseline treatment	0.216	(11.887)	1.939	(7.461)
Baseline × investment leader	−0.314*	(0.170)	−0.240**	(0.103)
Leading by example treatment	−22.999*	(11.955)	−11.961**	(8.140)
LbE × investment leader	0.203	(0.167)	0.099	(0.111)
Round t (1–20)			−0.422***	(0.122)
Investment t_{-1}			0.474***	(0.055)
Constant	21.326*	(11.673)	15.197*	(8.186)
R^2	0.510		0.626	
N (obs/matching group)	1280/32		1216/32	

Matching group included four groups that were randomly matched into contests across the rounds. In the second, each group constituted a matching group because the experiment does not allow for interaction between groups. ***: $p < .01$; **: $p < .05$; *: $p < .1$.

such as the content analysis of communication (e.g. Brandts & Cooper, 2007; Cason, Sheremeta, & Zhang, 2012; Eisenkopf, 2014) or the identification of emotions (Coan & Allen, 2007; Kugler, Ye, Motro, & Noussair, forthcoming).

It is also interesting that the two experiments provide similar qualitative results (with different levels of significance) regarding the leader-followership relationship even though the experimental games differ in the incentive structure and actual leadership behavior. This pattern may be down to coincidence. However, it suggests that key elements in the leader-follower relationship do not vary across different social dilemma situations. However, such a comparison requires a much more dedicated research design.

Practical implications

It is a popular line that politicians and managers should “walk the talk” and “practice what they preach”. Of course the promotion of honesty can be valuable in itself and the results of the competitive show that the alignment of words and deeds induces followers to act like their leader. The results thus support the relevance of behavioral integrity in leader decisions at least in competitive environments. However the experimental results also suggest that the economic benefits of revealed integrity are limited. Leaders make themselves vulnerable to exploitation by the other group members if they lead by example. Walking the talk requires courage and trust in reciprocal inclinations of the group. Hence, a group can become worse of if the leader reveals timidity with her decisions.

Nevertheless, real world leaders may have a greater need for behavioral integrity than randomly appointed experimental leaders. Many operate in low trust environments. Leadership appointments in particular often involve competitive processes that invite unfair behavior (Chen, 2003). Hence, followers may perceive their leader as less upright than an arbitrary person.

Conclusions

The paper provides an innovative contribution to the understanding of the impact of leading by example. The experimental approach complements “real world” studies and addresses key problems of causal identification. In particular, it disentangles the impact of words and deeds and provides a differentiated perspective on the interaction between leader and followers. The paper also shows that at least some of the more complex management concepts can be linked to more parsimonious theoretical contributions from behavioral economics. Thus experimental paradigms can provide substantial contributions for the study of leadership. However, the rather small size of some effects suggests that some enthusiastic proponents of specific leadership concepts should expect a certain disillusion. As in this case of behavioral integrity, cleaner evidence suggests more complex leader-follower relationships.

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Urs Fischbacher and my former colleagues at the Thurgau Institute of Economics supported the planning of this project with multiple questions and suggestions. Seminar participants at the Universities of Osnabrück and Vechta and at the Annual Meeting of the German Economic Association for Business Administration 2018 in Frankfurt gave me stimulating feedback. This gratitude also extends to the responsible editor and two referees who gave me detailed advice on how to address the wider audience of leadership researchers and practitioners. Debora Gehring and the research assistants of the Lakelab at the University of Konstanz provided excellent support by conducting the experiment. I gratefully acknowledge financial support for this project from the Deutsche Forschungsgemeinschaft (Grant EI 983/3-3). All errors remain my own.

Appendix A. Instructions for Experiment I

These instructions have been translated from German. All subjects in a treatment received the same instructions. Differences between the instructions across the treatments are highlighted. Numerical examples and On-Screen control questions are not included. The experiment started once all subjects had answered all control questions correctly.

Welcome to this economic experiment.

Your decisions and the decisions of the other participants will affect your payoff. Hence, it is important that you read these instructions carefully. Please contact us before the experiment starts if you have any question.

Please do not talk with the other participants during the experiment.

Otherwise we might exclude you from the experiment and any subsequent payment.

During the experiments we always talk about points that determine your income. At the end of the experiment we convert all points into Euros, using the following exchange rate.

120 points = 1 Euro

You get your payments at the end of the experiment in cash. Now we explain you the experiment in detail.

A.1. Experimental setup

At the start of the experiment you will form a group with two randomly chosen participants. There are three team members, A, B, and C. Participant C has a special role which we describe below. You will learn at the start of the experiment whether you are A, B, or C. During the experiment your group will meet another group of similar composition. In each you will meet another randomly chosen group.

The experiment has 20 rounds. The composition of your group will not change over these rounds. In each round, your group or the other group can get a prize of 300 points. The probability of success of your group depends on the inputs of its members. Each group member gets 100 points irrespective of the personal input if the group gets the prize.

At the start of each round each participant gets 100 points. Any participant can use these points as inputs (from 0 to 100). All points not invested will be added to the participant's private account.

Once the participants have decided about their inputs, these investments will be summed up. The success probability of your group depends on the ratio of its investments relative to the investments of the other group.

The computer adds up the inputs within a team. The success probability is derived from the ratio between your team's input and the sum of both teams' inputs. If both teams invest the same amount the success probability is 50% for each team. This also holds if both teams invest 0 points. If one team makes a higher investment the success probability is also higher. However, it is not guaranteed that the team with the higher investment also wins the prize. More specifically the formula for the success probability is as follows:

$$\text{Success probability} = \frac{\text{Input of your team}}{\text{Input of your team} + \text{Input of the other team}}$$

The income of any group member is determined as follows:

Payoff from the private account (= 100 − Investment) *+ 100 points (if the group wins the prize.)* *= total payoff*

A.2. Participant C

Participant C has a special role in the team.

Baseline treatment	Recommendation treatment	Learning-by-example treatment
–	Participant C can suggest the other group members how many points to use as input. Both A and B get the same recommendation. A and B can never exchange information among themselves.	
–	–	Player C then decides about her own input. Participants A and B learn about this input before they make their own investment decision.
At the end of a round each participant receives information about whether the own group has received the 300 points. Only participant C learns about the actual inputs of A and B and the aggregate inputs of the other group.		
- Numerical Examples -		

A.3. Timeline

One round proceeds as follows

1. You get 100 points.
2. *Recommendations* and *LbE Treatments*: Input recommendation by participant C.
3. *LbE Treatment only*: Input decision of participant C. The invested points will be withdrawn from the 100 point (only in)
4. Input decisions

- *Baseline* and *Recommendation Treatments*: All participants decide about their inputs. The invested points will be withdrawn from the 100 point
- *LbE Treatment*: The participants A and B learn about the input of C. They decide about their inputs. The invested points will be withdrawn from the 100 point

5. Distribution of the Prize. Participant C learns about the actual inputs of A and B and the aggregate inputs of the other group.

The experiment has 20 rounds that all proceed as mentioned above. You will remain in the same group throughout the entire experiment. We will add up all earned points, exchange them into Euro and pay them to you at the end of the experiment in cash.

Appendix B. Instructions for Experiment II

These instructions have been translated from German. All subjects in a treatment received the same instructions. Differences between the instructions across the treatments are highlighted. Numerical examples and On-Screen control questions are not included. The experiment started once all subjects had answered all control questions correctly.

Welcome to this economic experiment.

Your decisions and the decisions of the other participants will affect your payoff. Hence, it is important that you read these instructions carefully. Please contact us before the experiment starts if you have any question.

Please do not talk with the other participants during the experiment.

Otherwise we might exclude you from the experiment and any subsequent payment.

During the experiments we always talk about points that determine your income. At the end of the experiment we convert all points into Euros, using the following exchange rate.

120 points = 1 Euro

You get your payments at the end of the experiment in cash. Now we explain you the experiment in detail.

B.1. Experimental setup

At the start of the experiment you will form a group with two randomly chosen participants. There are three team members, A, B, and C. Participant C has a special role which we describe below. You will learn at the start of the experiment whether you are A, B, or C. The experiment has 20 rounds.

At the start of each round each participant gets 100 points. Any participant can use these points as inputs (from 0 to 100). All points not invested will be added to the participant's private account.

Once the participants have decided about their inputs, these investments will be summed up. The lab manager increases this sum by 50% and distributes all these points equally among the group members.

All group members benefit in the same way from your investment as you benefit from the inputs of the other group members. The income of any group member is determined as follows:

Payoff from the private account (= 100 − Investment)
+payoff from the inputs (= 0,5 × sum of all inputs)
=total payoff

B.2. Participant C

Participant C has a special role in the Team.

Baseline treatment	Recommendation treatment	Learning-by-example treatment
–	Participant C can suggest the other group members how many points to use as input. Both A and B get the same recommendation. A and B can never exchange information among themselves.	
–	–	Player C then decides about her own input. Participants A and B learn about this input before they make their own investment decision.
At the end of a round each participant receives information about the own payoff. Only participant C learns about the actual inputs of A and B.		

- Numerical Examples -

B.3. Timeline

One round proceeds as follows

1. You get 100 points.
2. *Recommendations* and *LbE Treatments*: Input recommendation by participant C.
3. *LbE Treatment only*: Input decision of participant C. The invested points will be withdrawn from the 100 point (only in)
4. Input decisions

a. *Baseline* and *Recommendation Treatments*: All participants decide about their inputs. The invested points will be withdrawn from the 100 point
b. *LbE Treatment*: The participants A and B learn about the input of C. They decide about their inputs. The invested points will be withdrawn from the 100 point

5. Distribution of the payments. Participant C learns about the actual inputs of A and B.

The experiment has 20 rounds that all proceed as mentioned above. You will remain in the same group throughout the entire experiment. We will add up all earned points, exchange them into Euro and pay them to you at the end of the experiment in cash.

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